

University of Gujrat

Department of Software Engineering

Object Oriented programming (Lab)

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AB

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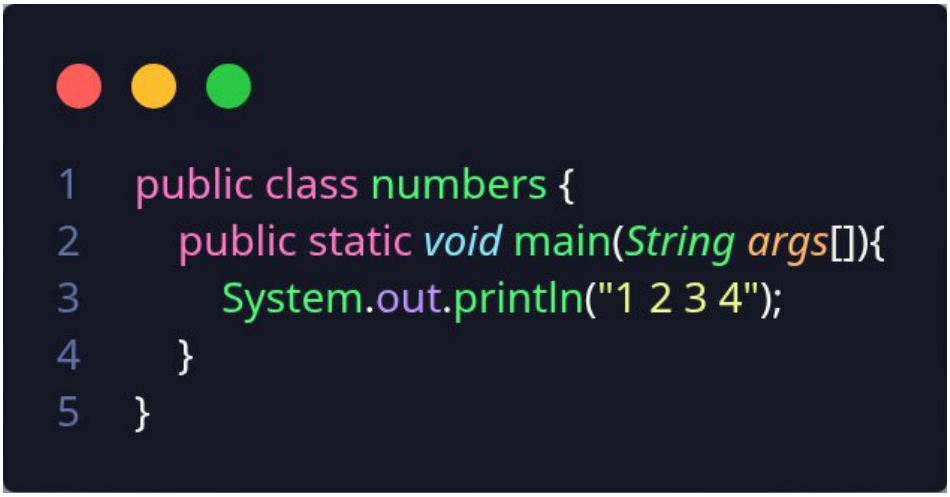
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Lab 1

- Write a program that displays the number 1 to 4 on the same line with each pair of adjacent numbers separated by space using a “System.out.print” Statement.



```
1  public class numbers {
2      public static void main(String args[]){
3          System.out.println("1 2 3 4");
4      }
5  }
```

Lab 1

- Write a program that asks the user to enter 2 integers and print the square of each, the sum of their square and the difference of their square

```java
import java.util.Scanner;
public class sqrSumDiff {
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter the first number:");
        int num1 = sc.nextInt();
        System.out.println("Enter the second number:");
        int num2 = sc.nextInt();
        int sq1=num1*num1;
        int sq2=num2*num2;
        int sqrsum = sq1 + sq2;
        int sqrdiff = sq1-sq2;
        System.out.println("The square of first number is:" + sq1);
        System.out.println("The square of second number is:" + sq2);
        System.out.println("The sum of squares is: " + sqrsum);
        System.out.println("The difference of squares is: " + sqrdiff);
    }
}

```

Lab 1

- Write a program that inputs 3 integers from users and displays the sum, average, product, the smallest and the largest of numbers.

```java
import java.util.Scanner;
public class aveProSmlLrg {
public static void main(String[] args) {
Scanner sc=new Scanner(System.in);
System.out.println("Enter the first number");
int num1=sc.nextInt();
System.out.println("Enter the second number");
int num2=sc.nextInt();
int num3=sc.nextInt();
int sum=num1+num2+num3;
int average=sum/3;
int product=num1*num2*num3;
int smallest=num1;
if(num2<smallest){
smallest=num2;
}
if(num3<smallest){
smallest=num3;
}
int largest=num1;
if(num2>largest){
largest=num2;
}
if(num3>largest){
largest=num3;
}
System.out.println("Sum: " + sum);
System.out.println("Average: " + average);
System.out.println("Product: " + product);
System.out.println("Smallest: " + smallest);
System.out.println("Largest: " + largest);
}

}
```

Lab 1

- Write a programs that takes one integer from user an an input and displays whether the number and its square are greater than, equal to or lesser than 1000.

```java
import java.util.Scanner;
public class lessGreat1000 {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.println("Please Enter a number");
int num =sc.nextInt();
if(num>1000){
System.out.println("The number is greater then 1000");
} else if(num==1000){
System.out.println("The number is equal to 1000");
} else{
System.out.println("The number is not greater than 1000");
}
if(num*num>1000){
System.out.println("The square of the number is greater then 1000");
} else if(num*num==1000){
System.out.println("The square of the number is equal to 1000");
} else{
System.out.println("The square of the number is not greater than 1000");
}
}
}
```

Lab 2

- Write a Java program that takes an integer input from the user and prints a message only if the number is positive.

```java
import java.util.Scanner;
public class positive {
public static void main(String[] args) {
Scanner sc= new Scanner(System.in);
System.out.println("Enter a number:");
int num=sc.nextInt();
if(num>0){
System.out.println("Good to have some positivity in your life");
}
}
}
```

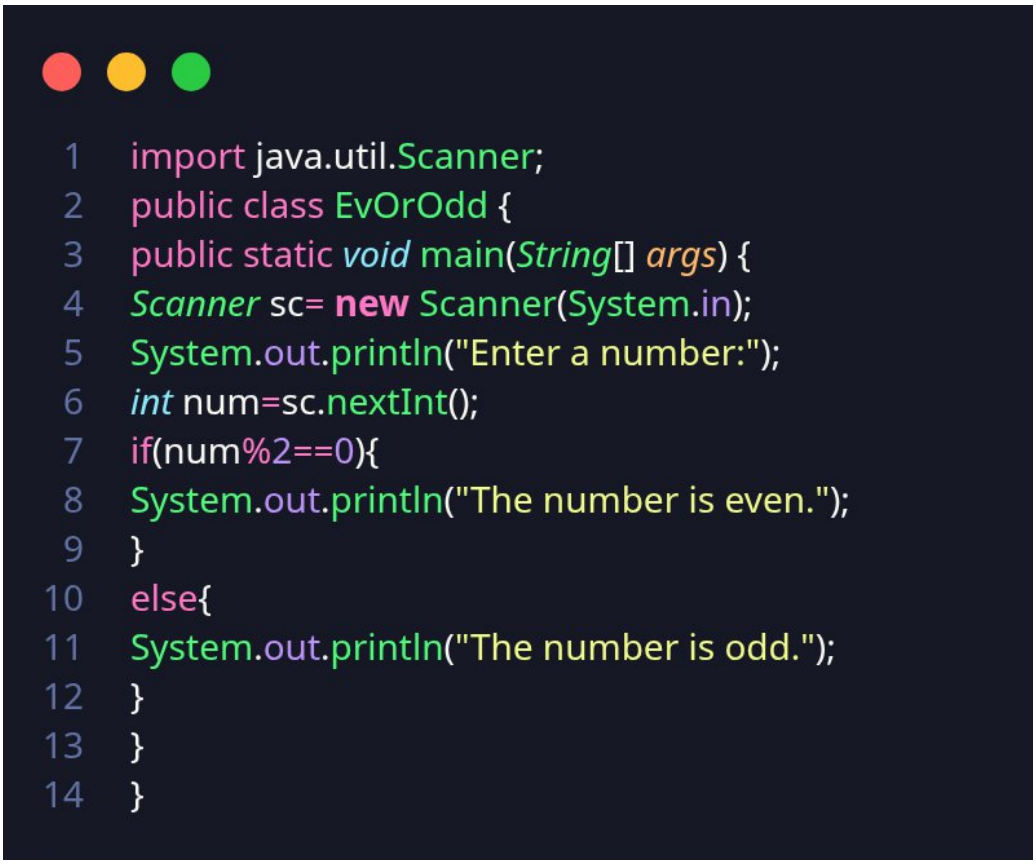
- Write a Java program that takes marks as input and prints "You passed the exam" only if marks are greater than or equal to 50

```java
import java.util.Scanner;
public class pass {
public static void main(String[] args) {
Scanner sc= new Scanner(System.in);
System.out.println("Enter your marks:");
int marks=sc.nextInt();
if(marks>=50){
System.out.println("Yeah. You passed the exam.");
}
}
}

```

Lab 2

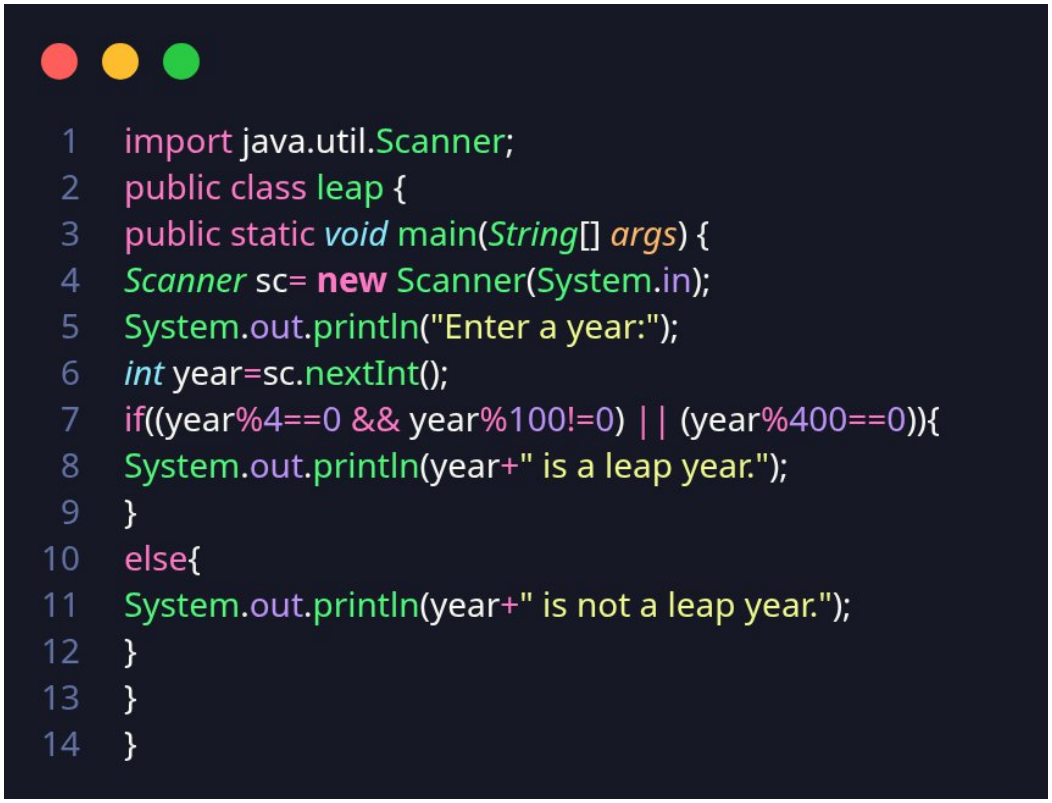
- Write a Java program that takes an integer from the user and checks whether the number is even or odd.



```java
import java.util.Scanner;
public class EvOrOdd {
public static void main(String[] args) {
Scanner sc= new Scanner(System.in);
System.out.println("Enter a number:");
int num=sc.nextInt();
if(num%2==0){
System.out.println("The number is even.");
}
else{
System.out.println("The number is odd.");
}
}
}
```

Lab 2

- Write a Java program that checks whether a given year is a leap year or not.



```java
import java.util.Scanner;
public class leap {
public static void main(String[] args) {
Scanner sc= new Scanner(System.in);
System.out.println("Enter a year:");
int year=sc.nextInt();
if((year%4==0 && year%100!=0) || (year%400==0)){
System.out.println(year+" is a leap year.");
}
else{
System.out.println(year+" is not a leap year.");
}
}
}
```

Lab 2

- Write a Java program to calculate electricity bill based on units consumed: If units ≤ 100 charge Rs. 5 per unit Otherwise → charge Rs. 10 per unit

```java
import java.util.Scanner;
public class bill {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.println("Enter the number of units consume
d:");int units = sc.nextInt();
int bill;
if (units <= 100) {
bill = units * 5;
} else {
bill = units * 10;
}
System.out.println("The electricity bill is: Rs. " + bill);
}
}
```

Lab 2

- Write a Java program that takes three numbers from the user and prints the largest number.

```java
import java.util.Scanner;
public class largest {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.println("Enter three numbers:");
int num1 = sc.nextInt();
int num2 = sc.nextInt();
int num3 = sc.nextInt();
int largest;
if (num1 >= num2 && num1 >= num3) {
largest = num1;
} else if (num2 >= num1 && num2 >= num3) {
largest = num2;
} else {
largest = num3;
}
System.out.println("The largest number is: " + largest);
}
}
```

Lab 2

- Write a Java program that takes marks as input and prints the grade:
  - Marks ≥ 90 → Grade A
  - Marks ≥ 75 → Grade B
  - Marks ≥ 50 → Grade C
  - Marks < 50 → Grade F

```java
import java.util.Scanner;
public class result {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.println("Enter the marks:");
int marks = sc.nextInt();
char grade;
if (marks >= 90) {
grade = 'A';
} else if (marks >= 75) {
grade = 'B';
} else if (marks >= 50) {
grade = 'C';
} else {
grade = 'F';
}
System.out.println("The grade is: " + grade);
}
}
```

Lab 2

- Write a Java program that takes a year as input and checks whether it is a leap year or not. A year divisible by 400 → Leap year. A year divisible by 100 but not 400 → Not a leap year. Otherwise → Not a leap year

```java
import java.util.Scanner;
public class leapS {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.println("Enter a year:");
int year = sc.nextInt();
boolean isLeapYear;
if (year % 400 == 0) {
isLeapYear = true;
} else if (year % 100 == 0) {
isLeapYear = false;
} else if (year % 4 == 0) {
isLeapYear = true;
} else {
isLeapYear = false;
}
if (isLeapYear) {
System.out.println(year + " is a leap year.");
} else {
System.out.println(year + " is not a leap year.");
}
}
}
```

Lab 2

- Write a Java program that asks the user for their age:
  - Age < 18 → "Not eligible to vote"
  - Age ≥ 18 → "Eligible to vote"
  - If Age ≥ 60 → Also print "You are a senior citizen

```java
import java.util.Scanner;
public class vote {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.println("Enter your age:");
int age = sc.nextInt();
if (age < 18) {
System.out.println("Not eligible to vote");
} else {
System.out.println("Eligible to vote");
if (age >= 60) {
System.out.println("You are a senior citizen");
}
}
}
}
```

Lab 2

- Write a Java program that takes two numbers and an operator (+, -, *, /) as input and performs the operation using a switch statement.

```java
import java.util.Scanner;
public class switchArithmetics {
public static void main(String[] args) {
Scanner sc= new Scanner(System.in);

System.out.print("Enter first number: ");
double num1 = sc.nextDouble();

System.out.print("Enter second number: ");
double num2 = sc.nextDouble();

System.out.print("Enter operator (+, -, *, /): ");
String op = sc.next();

double result;
switch (op) {
case "+":
result = num1 + num2;
System.out.println("Result: " + result);
break;
case "-":
result = num1 - num2;
System.out.println("Result: " + result);
break;
case "*":
result = num1 * num2;
System.out.println("Result: " + result);
break;
case "/":
if (num2 == 0) {
System.out.println("Division by zero is not allowed.");
break;
}
result = num1 / num2;
System.out.println("Result: " + result);
break;
default:
System.out.println("Wrong operator. Please use +, -, *, or /.");
}
}
}
```

Lab 2

- Write a Java program that takes a month number (1–12) as input and prints the number of days in that month. For February, assume 28 days.

```java
import java.util.Scanner;

public class daysInMonth {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);

System.out.print("Enter month number (1-12): ");
int month = sc.nextInt();

int days;
switch (month) {
case 1:
days = 31;
break;
case 2:
days = 28;
break;
case 3:
days = 31;
break;
case 4:
days = 30;
break;
case 5:
days = 31;
break;
case 6:
days = 30;
break;
case 7:
days = 31;
break;
case 8:
days = 31;
break;
case 9:
days = 30;
break;
case 10:
days = 31;
break;
case 11:
days = 30;
break;
case 12:
days = 31;
break;
default:
System.out.println("Invalid month. Please enter a number from 1 to 12.");
return;
}

System.out.println("Number of days: " + days);
}
}
```